

Counterfactuals, Laws, and History

Topics in Language and Mind

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1. Holding History Fixed

Claim: usually, when we evaluate an ordinary counterfactual ‘If it were that A it would be that C’, where A is concerned with a particular interval of time, we tacitly “hold fixed” a very broad range of propositions about history before that interval, in the sense that we assume that if these propositions are in fact true, they would also have been true if A had been true.

Illustrations:

- (1) If I had done such-and-such, I would have revenged a wrong done me last year.
- (2) If John had skipped breakfast today, that would have been the first time he did so in months.
- (3) If this video were downloaded 10,000 times, that would set a new record for us.

However, as Lewis recognises, there are some circumstances where we do not reason like this. Contrast:

- (4) If I had jumped out of that tenth-floor window, I would have been killed.
- (5) If I had jumped out of that tenth-floor window, that would have to have been because someone had a safety net in place. (Jackson)

These two sentences naturally suggest two *different contexts*. If they were both true in the same context,

- (6) If I had jumped out of that tenth-floor window, I would have been killed and there would have been a safety net in place.

would also have to be true in that context; but this doesn’t seem to be true in any normal context. Lewis calls the kind of context evoked by (5) a “backtracking context”, and the kind of context evoked by (4) a “standard context”.

Lewis’s project: give an analysis that captures the truth conditions of ‘If it were that A, it would be that C’ in a *standard* context.

- Note that one might think there was further important context-sensitivity *within* the set of standard contexts. (Come back to this.)

2. An account not in terms of closeness

Assuming we accept Lewis’s logic for counterfactuals, there is good reason to think that any possible reading a counterfactual *can* be captured by some combination of a closeness relation and an accessibility relation. But this does not mean we are *forced* to

state the truth conditions in this way! Lewis begins with an analysis, 'Analysis 1', that is not expressed in terms of closeness / accessibility:

ANALYSIS 1. Consider a counterfactual "If it were that A, then it would be that C", where A is entirely about affairs in a stretch of time t_A . Consider all those possible worlds w such that:

- (1) A is true at w ;
- (2) w is exactly like our actual world at all times before a transition period beginning shortly before t_A ;
- (3) w conforms to the actual laws of nature at all times after t_A , and
- (4) during t_A and the preceding transition period, w differs no more from our actual world than it must to permit A to hold.

The counterfactual is true if and only if C holds at every such world w .

Lewis thinks that Analysis generally assigns the right truth values to counterfactuals (understood relative to a standard context). His complaints are as follows:

- It is too weak - it is just silent about the truth conditions of counterfactuals whose antecedents are not entirely about affairs in a stretch of time.
- It incorrectly rules out backwards counterfactual dependence (outside the 'transition period') as *absolutely* impossible, even under weird laws of nature.

Further reasons to continue the inquiry: Analysis 1 does not fit into the general closeness framework (or any other general framework that deals with all contexts, not just the standard ones); and it leaves us in need of an explanation of why we have *this* concept, rather than a different concept in which the roles of 'past' and 'future' are reversed.

3. Determinism

Determinism: all truths follow with metaphysical necessity from (i) the laws of nature, together with (ii) the complete intrinsic truth about any initial segment of the history of the universe.

4. A controversial feature that Analysis 1 shares with Lewis's final view: miracles

Suppose determinism is true. Both of the following claims seem tempting:

Laws: If you hadn't come to class today, every actual law of nature would have been true.

Past: If you hadn't come to class today, the intrinsic character of the first billion-year stretch of the history of the universe since the Big Bang would have been exactly the same.

But assuming determinism, the actual laws plus the intrinsic truth about that interval jointly entail that you do come to class today. Given the principle that $P > Q$ entails $P >$

R whenever P is metaphysically possible and Q entails R, this means that if we accept both *Laws* and *Past* we have to say that if you hadn't come to class today you would have come to class today, which is absurd. So, either *Laws* or *Past* have to be given up.

Lewis gives up *Laws*: he thinks that assuming determinism is true, if things had been different at one time, a 'miracle' would have occurred shortly before that time.

NB: That does not mean that false propositions can be laws. The miracle is inconsistent with the *actual* laws, not with anything that would have been a law if you hadn't come to class.

An alternative approach: keep *Laws* and reject *Past*, even for ordinary ('non-backtracking' contexts).

Does that mean just rejecting the asymmetry of counterfactuals altogether? Not necessarily. We could say that if things had been different at *t*, the course of history before *t* would have been only *slightly* different, whereas the course of history after *t* would have been significantly different.

Lewis worries about this: 'Indeed, it is hard to imagine how two deterministic worlds anything like ours could possibly remain just a little bit different for very long. There are altogether too many opportunities for little differences to give rise to big differences.' (45).

- In 'Against Counterfactual Miracles', section 4, I argue that there is nothing to worry about here.

Q: how could Analysis 1 be revised to fit with the law-preserving approach?

Q: How could Lewis's final analysis be revised to fit with the law-preserving approach?

5. Another controversial feature that Analysis 1 shares with Lewis's final view: the "transition period"

Lewis does not think that if things had been intrinsically different at *t* the *entire* past of *t* would have been *exactly* the same—he allows for a 'transition period'.

What motivates this?